



R3.3 Electron sharing reactions

Reactivity 3 · What are the mechanisms of chemical change? · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Radical substitution occurs when alkanes react with halogens under UV light, proceeding via a chain mechanism: initiation (UV light homolytically breaks the halogen-halogen bond producing two radicals), propagation (radicals react with molecules to produce a new radical and a new product, in a repeating cycle that sustains the chain), and termination (two radicals combine to form a stable molecule, ending that chain). Alkenes undergo electrophilic addition reactions because their C=C double bond has a region of high electron density (the π bond) that attracts electrophiles; for example, addition of a hydrogen halide or halogen across the double bond proceeds via a carbocation intermediate (for asymmetric alkenes with unsymmetrical reagents, Markovnikov's rule predicts the major product forms via the more stable carbocation).

Worked example: follow the reasoning

Outline the mechanism for the free-radical substitution of methane by chlorine in the presence of UV light, to form chloromethane.

Step 1 (Initiation): UV light provides energy to homolytically break the Cl-Cl bond: $\text{Cl}_2 \rightarrow \text{Cl}\cdot + \text{Cl}\cdot$ (each Cl atom keeps one electron from the shared pair).

Step 2 (Propagation, step 1): A chlorine radical collides with methane and abstracts a hydrogen atom: $\text{Cl}\cdot + \text{CH}_4 \rightarrow \cdot\text{CH}_3 + \text{HCl}$.

Step 3 (Propagation, step 2): The methyl radical reacts with another Cl_2 molecule: $\cdot\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$. This regenerated $\text{Cl}\cdot$ radical continues the chain.

Step 4 (Termination): Two radicals combine, e.g. $\cdot\text{CH}_3 + \text{Cl}\cdot \rightarrow \text{CH}_3\text{Cl}$, or $\text{Cl}\cdot + \text{Cl}\cdot \rightarrow \text{Cl}_2$, or $\cdot\text{CH}_3 + \cdot\text{CH}_3 \rightarrow \text{C}_2\text{H}_6$, removing radicals and ending the chain.

Answer: The overall reaction is $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$, proceeding via the radical chain mechanism of initiation, propagation and termination described above.

Quick check · multiple choice

When HBr is added to propene, $\text{CH}_3\text{CH}=\text{CH}_2$, the major product is 2-bromopropane rather than 1-bromopropane. Which statement best explains this (Markovnikov's rule)?

- A The reaction proceeds via a more stable secondary carbocation intermediate
- B The reaction proceeds via a less stable primary carbocation intermediate
- C Bromine always adds to the terminal (least substituted) carbon
- D The mechanism involves a free-radical intermediate, not a carbocation

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Ethene reacts with bromine, Br_2 , in an electrophilic addition reaction. [8]
 - (a) Write the overall equation for this reaction and name the organic product. [2]
 - (b) Explain why the C=C double bond in ethene is susceptible to attack by an electrophile such as Br_2 , despite Br_2 itself being non-polar. [3]
 - (c) Outline the mechanism of this reaction, including how the non-polar Br_2 molecule becomes polarised as it approaches the alkene. [3]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $\text{CH}_2=\text{CH}_2 + \text{Br}_2 \rightarrow \text{CH}_2\text{BrCH}_2\text{Br}$. Product: 1,2-dibromoethane. [2]

(b) The $\text{C}=\text{C}$ double bond contains a π bond with electron density concentrated above and below the plane of the molecule; this creates a region of high electron density that can repel/induce a dipole in an approaching non-polar molecule and then attract/react with an electrophile (electron-pair acceptor), even though Br_2 itself has no permanent dipole. [3]

(c) As the Br_2 molecule approaches the electron-rich $\text{C}=\text{C}$ double bond, the high electron density repels the electrons in the $\text{Br}-\text{Br}$ bond, inducing a temporary dipole ($\text{Br}^{\delta+}-\text{Br}^{\delta-}$); the π electrons of the double bond then attack the $\delta+$ bromine atom, forming a new $\text{C}-\text{Br}$ bond and breaking the $\text{Br}-\text{Br}$ bond heterolytically (the bromide ion Br^- leaves), generating a carbocation intermediate on the other carbon; the Br^- ion then attacks this carbocation, forming the second $\text{C}-\text{Br}$ bond and giving the dibromo product. [3]

Quick check answer

A. Protonation of the double bond occurs to form the more stable secondary carbocation (rather than the less stable primary carbocation), and Br^- then adds to this carbon, giving 2-bromopropane as the major product.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.